Voice Controlled Smart Wheel Chair With GPS Tracking

INNOVATIVE PRODUCT DEVELOPMENT - IV REPORT

Submitted by

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Under the Esteemed Mentorship of

(R.Durga Bhavani)

(Assistant Professor)

in partial fulfillment of the Academic Requirements for the Degree of

BACHELOR OF TECHNOLOGY

Electronics and Communication Engineering

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)

Accredited by NAAC with 'A+' Grade | Programmes Accredited by NBA

NIRF National Institutional Ranking 2025 – Rank band(201-300), MHRD, Govt. of India

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the Innovative Product Development – IV work entitled "**Voice Controlled Smart Wheel Chair With GPS Tracking**" is carried out by **B. MANEESHA (23RH1A0415) , CH. SARASWATHI(23RH1A0426) , K. DIVYA(23RH1A0455) of III B.Tech – II Sem** in partial fulfillment for the award of degree of **BACHELOR OF TECHNOLOGY** in Electronics and communication Engineering, Malla Reddy Engineering College of Women - Hyderabad during the academic year 2025-2026.

Mentor Signature
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External Examiner

ACKNOWLEDGEMENT

We feel ourselves honored and privileged to place our warm salutation to our college **Malla Reddy Engineering College for Women** and Department of **Electronics and Communication Engineering** which gave us the opportunity to have expertise in engineering and profound technical knowledge.

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With Regards and Gratitude
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DECLARATION

We hereby declare that our Innovative Product Development-IV entitled **"Voice Controlled Smart Wheel Chair With GPS Tracking"** submitted to **Malla Reddy Engineering College for Women,** affiliated to **Jawaharlal Nehru Technological University, Hyderabad** for the award of the Degree of Bachelor of Technology in **Electronics and Communication Engineering** is a result of original research work done by us.

It is declared that the Innovative Product Development-IV report or any part there of has not been previously submitted to any University or Institute for the award of Degree.

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VOICE CONTROLLED SMART WHEEL CHAIR WITH GPS TRACKING

ABSTRACT

This project is related to controlling a wheel chair by means of human voice. The objective of this project is to facilitate the movement of people who are disabled or handicapped. Speech recognition technology is used to move the wheel chair. The result of this project is show that this can be used for future research work and public interest. The Voice Controlled Smart Wheelchair with GPS Tracker is an innovative assistive device designed to help physically disabled and elderly individuals move independently. The system uses voice recognition technology to control the movement of the wheelchair, reducing the need for manual operation. A GPS tracking module is integrated to monitor the real-time location of the wheelchair, ensuring safety and easy tracking in case of emergencies. The system improves mobility, safety, and convenience, making it a valuable solution in modern healthcare.

CHAPTER-1

INTRODUCTION

This project aims to style integrate program to interface and test a totally motorized, voice operated wheelchair. In this project, the procedure of the Mechatronic systems design was followed to assure the quality of the final product i.e; the Voice controlled wheelchair. The project took the following of the subsequent parts: Hardware, software,
interface, and testing.

The target of this project is to facilitate the movement of people who are disabled or handicapped. We used Speech recognition technology which has been employed to maneuver the wheelchair. The results of this project show that this will be used for future research work and public interest. The main objective is to style a system that gives the answer for the physically handicapped (challenged) people those who that can't move by themselves, they can use speech commands by interfacing the Speech Recognition with a microcontroller and wheelchair. The voice commands are given to the Bluetooth module HC-05 kit with the help of an android phone and thus the wheelchair moves according to the given directions. The wheelchair's movement is being controlled by the motors and the motor drivers are being connected to the wheels of the chair The interfacing between speech recognition kit and motors is completed by employing a microcontroller.

The interfacing between speech recognition kit and motors is completed by employing a microcontroller. Here during this project, the microcontroller used is arudino The idea was taken during this paper to reduce the human efforts in driving a wheelchair. It's an incontrovertible fact that humans are born imperfections. But disabilities have never stopped a person from achieving greatness. Furthermore, many devices/instruments are invented to assist those that are mentally or physically challenged. According to the earliest records, the wheeled furniture is an inscription that has been found on a stone slate in China and as a child's bed depicted during a frieze on a Greek vessel, both beginnings in the 6th and 5th century BCE.

CHAPTER 2

EMBEDDED SYSTEM

2.1 Embedded System

A schema is a strategy for operational, dealing with or doing one or various endeavors according to a settled orchestrate, package, or set of rules. A skeleton is moreover preparation its chip in as shown by the game plan or venture. For example, consider a watch. It is a period show schema. Its parts are its supplies, needles and battery with the radiant dial, body and strap. These parts make to show the nonstop reliably and continually overhaul the time reliably. The schema framework overhauls the presentation using three needles after consistently. It takes after a set of principles. An inserted framework may be a framework that goes to attempt and do a unimportant assignment is that the connected outline and this level delineated as combo of every machine code and fittings. Generally handy importance of monitor or help the procedure of mechanical assembly, hardware or plant. "Installed" the very truth that partner degree inverse compelling a broadly functional smart phone could likewise be wont to administration the operation of an oversized muddled procedure plant, and its vicinity is selfevident.

All embedded systems square measure as well as computers or microprocessors. A number of these laptops square measure but terribly easy classifications related through private processer. The unpleasantly square measure expressions singularly one performs or set of capacities to fulfill one arranged reason. In extra confounded frameworks partner degree machine developer that permits the installed structure to be used for a picked reason in the midst of a specific application chooses the working of the embedded structure. This is versatility to claim projects imply a near embedded schema is used for an extent of diverse functions. Now and again a microchip could likewise be composed in such somehow that application machine code for a chose reason for existing is quality added to the fundamental machine code amid a second technique, when that its unendurable to structure extra changes.

The most run contraption contains one microchip (habitually intimated as a "chip"), which itself can be prepackaged with unique chips in the midst of a crossbreed structure or application particular microcircuit. Its data starts from a locator or contraption and its yield goes to a switch or substance that could begin or stop the operation of a machine or, by operational a valve, could affiliation the stream of fuel to embellishment degree engine.

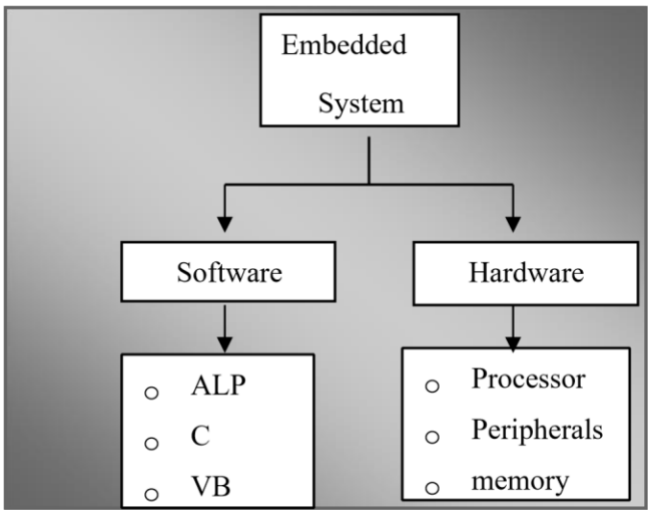


Fig.2.1: Block diagram of Embedded System

2.2 Classification of Embedded Systems

2.2.1 Small Scale Embedded Systems

The skeletons are delineated a single 8- or 16-bit microcontroller. The programming difficulties and incorporate sheet level design. Exactly when making embedded programming for these, an editor, developing executor and cross building operator, specific used, are the rule programming contraptions. By and large, "C" is used for making structures. "C" platform game plan is done into the social affair, also executable codes are then fittingly situated in the pattern memory. The thing needs to fit inside the memory open and keep in view the need to most inaccessible point power dispersal when framework is running on and on.

2.2.2 Medium Scale Embedded Systems

These skeletons are customarily made with a solitary or few 16- or 32-bit micro controllers or Dips or Reduced Instruction Set Computers (Risks). These have both apparatus and programming complexities. For complex programming chart, there are the running with programming mechanical gatherings Source code planner instrument, Simulator, debugger and incorporated improvement environment. Programming mechanical assemblies in like manner give the responses for the gear complexities. A developing executor is of little use as a programming instrument. These skeletons may furthermore use the instantly available Asps and Is (elucidated later) for the diverse limits case in point, for the vehicle interfacing, scrambling, translating, discrete cosine change and regressively transformation, TCP/IP tradition stacking and framework joining limits.

2.2.3. Sophisticated Embedded System

Forefront presented skeletons have colossal fittings and programming complexities and may require adaptable processors or configurable processors and programmable defense bunches. They are utilized for front line applications that need fittings and programming codesign and trade off in continue going skeleton; on the other hand, they are obliged by the arranging paces open in their supplies units. Certain thing points of confinement, case in point, encryption and unraveling reckonings, discrete cosine change and inverse change estimations, TCP/IP custom stacking and system driver breaking points are executed in the fittings to get extra speeds by sparing time. A package of the points of confinement of the apparatus assets in the structure are besides acknowledged by the thing. Change mechanical assemblies for these schemas may not be instantly available at a sensible cost or may not be open at all.

2.3 Memory Architecture

There two diverse sort's memory architectures there are:

- Harvard Architecture
- Von-Neumann Architecture

2.3.1 Harvard Architecture

Workstations have separate memory regions for project guidelines and information. There are two or more inside information transports, which permit concurrent access to both directions and information. The CPU gets program directions on the system memory transpose.

The Harvard building design is workstation construction modeling through actually separate stockpiling indicator guidelines from the information. This started hand-off created workstation, put away directions information restricted information stockpiling, completely

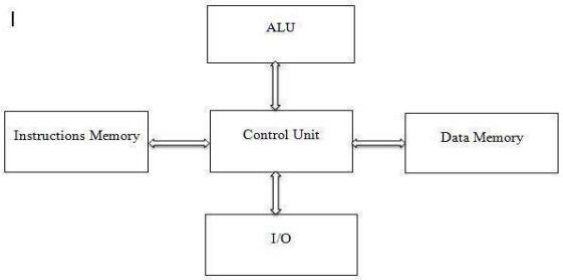


Fig.2.2: Harvard Architecture

2.3.2 Von-Neumann Architecture

A machine has a solitary, basic memory space in which both project guidelines and information are put away. There is a solitary inside information transport that gets both directions and information. They can't be performed in the meantime

The Von Neumann construction modeling is an outline model for a put away program computerized workstation that is use a focal preparing unit control processing unit and a solitary separate stockpiling structure ("memory") to clasp both guidelines and information. This is machine researcher machines actualize an all-inclusive Turing machine and have a consecutive structural engineering. The expressions "von Neumann structural engineering" and "put away program workstation" are for the most part utilized conversely, and that utilization is followed in this article.

The systems for exchanging the information and directions between the CPU and memory are, be that as it may, extensively more unpredictable than the first von Neumann

2.4 APPLICATIONS OF EMBEDDED SYSTEMS:

- Consumer applications:

At home we use a number of embedded systems which include microwave oven, remote control, vcd players, dvd players, camera etc…

- Industrial automation:

Today a lot of industries are using embedded systems for process control. In industries we design the embedded systems to perform a specific operation like monitoring temperature, pressure, humidity ,voltage, current etc.., and basing on these monitored levels we do control other devices, we can send information to a centralized monitoring station.In critical industries where human presence is avoided there we can use robots which are programmed to do a specific operation

CHAPTER 3

RELATED WORK

The other paper which we through focuses on the development of solar based wheelchair which helps in reducing the running cost of the wheelchair. It is specifically created for hemipelagic users to control the wheelchair efficiently. The control of the wheelchair can be assigned to either side to help the user. The wheelchair has significant benefits but still restricts from the circumstance that it has larger battery charging time and less capacity to climb up the slope

The other paper specified about the high costing of the electric wheelchair and modifying the present electric wheelchairs to make it affordable to the common people. It put importance on the utilization of sources open locally to cut the price of wheelchair. It utilizes pair of dc gear motor, PIC microcontroller and h bridge module for commanding the movement of wheelchair. The shortcoming of this research is that expense of the wheelchair is still huge for the ordinary people to afford it.

After looking at all the work we came to the decision that there was no voice-controlled wheel chair which would use an android phone for controlling it through the App where it would send signal through Bluetooth to the HC05 and the rest of the function would take place. Hence our idea would be the first of its kind. But it came with the added disadvantage of 3rd party intervention and failure of the voice control leads to failure of the entire system.

CHAPTER -4

EXISTING SYSTEM

An obstacle-based wheelchair system is a smart mobility solution designed to enhance safety and convenience for individuals with physical disabilities. This system incorporates sensors such as ultrasonic, infrared, or laser-based detectors to identify obstacles in the wheelchair's path. When an obstacle is detected, the system automatically processes the data and either slows down, stops, or redirects the wheelchair to avoid collisions. The technology can be integrated with autonomous or semi-autonomous control systems, ensuring smooth and safe navigation, even in complex environments. Additionally, some systems allow for customization of the obstacle detection sensitivity based on user preferences or environmental conditions. By reducing the risk of accidents and enhancing user confidence, an obstacle-based wheelchair system greatly improves the independence and mobility of users.

DRAWBACKS

Limited Sensor Range

Sensitivity to Environmental Conditions

CHAPTER -5

PROPOSED SYSTEM

The purpose of our research is to provide simpler wheelchair hardware architecture but with powerful computational platforms so that wheelchairs or the robot's designer can focus on their research and tests instead of Bluetooth connection infrastructure. This simplistic architecture is also helpful for educational robotics, because students can construct their own robots at an economical cost and use them as platform for experiments in several courses. The main purpose of this project is to develop a remote user interface to control a system via a wireless technology. There is a need to communicate with the wheelchair remotely in order to control the wheelchair's movements and pass critical data both ways.

Block diagram

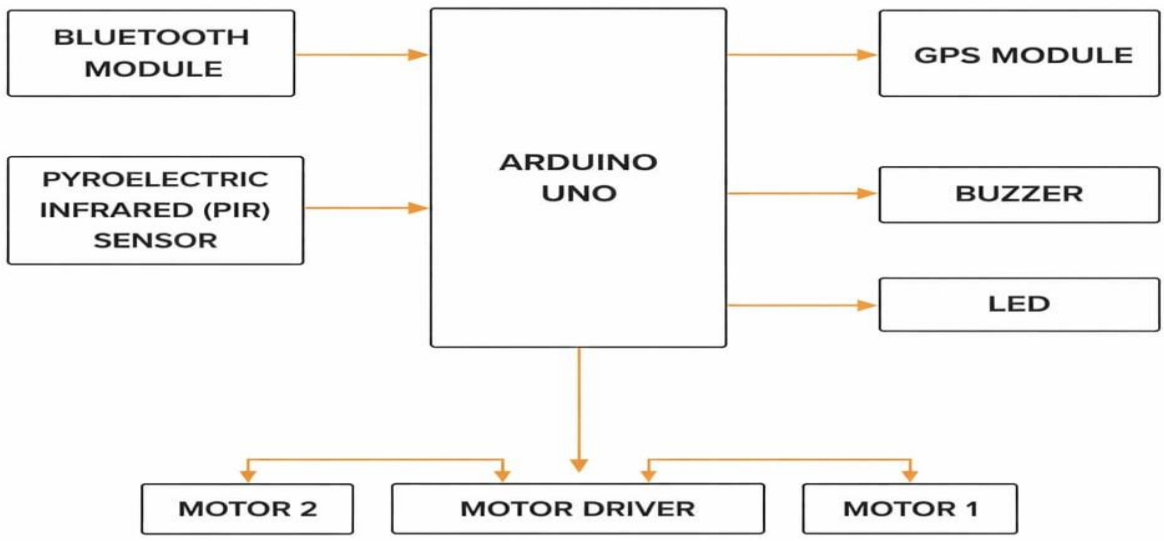


Fig: Proposed System

The hardware description section provides a detailed exposition of the key components driving our smart wheelchair. The Bluetooth module, operating within the 2.4 GHz frequency range, establishes secure wireless communication, enabling real-time data exchange and performance monitoring. The ultrasonic sensor, utilizing high-frequency sound waves, detects obstacles and provides crucial feedback for safe navigation. The motor driver (L293D) acts as the powerhouse, translating commands from the Arduino Uno microcontroller into precise motor movements, ensuring seamless mobility in all directions. A **GPS module** is connected to the microcontroller It continuously receives signals from satellites. The module calculates the **real-time location** of the wheelchair. This location data is sent to a **GSM module or mobile device**.

The wheelchair paths and progress probable are as given below.

1. **Forward:** Both motors are in forward direction.

2. **Left:** Left motor closed and right motor in front direction.

3. **Right:** Right motor stopped and left motor in forward direction.

4. **Stop:** Both motors are stopped.

Movement of the motors is controlled with the help of joystick commands and Arduino UNO and battery using joystick controller board. When the DC driver is connected to the motors and supply of 12v is given from the external voltage chargeable source. Interfacing of Arduino and HC-05 is done. With the help of android phone connect the Bluetooth HC05 to the android app it is an independent speaker app.

Forward: Motor1 moves in the anti-clockwise direction. Similarly, Motor2 moves in the anticlockwise direction and the tyres of the wheelchair moves in forward direction. **Left:** Motor1 or the left tyre moves in the anti-clockwise direction and simultaneously Motor2 or the right tyre moves in the clockwise direction.

Right: Motor2 or the right tyre moves in the anti-clockwise direction and simultaneously Motor1 or the left tyre moves in the clockwise direction

Stop: Both motors are stopped.

GPS Tracking

In the **Voice Controlled Smart Wheelchair Using GPS Tracker**, the GPS module plays a very important role in ensuring **safety and tracking** of the user.

A **GPS module (like Neo-6M)** is connected to the microcontroller (Arduino).

It continuously receives signals from satellites.

The module calculates the **real-time location (latitude & longitude)** of the wheelchair.

This location data is sent to a **GSM module or mobile device**.

Caregivers or family members can track the wheelchair location easily.

The integration of GPS in the **Voice Controlled Smart Wheelchair** enhances user safety by enabling real-time location tracking. It ensures that the user can be monitored at all times, making the system more reliable and suitable for practical use in daily life.

CHAPTER 6

ARDUINO CONTROLLER

The Arduino UNO is a widely used open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits The board features 14 Digital pins and 6 Analog pins. It is programmable with the Arduino IDE via a type B USB cable. It can be powered by a USB cable or by an external 9 volt battery, though it accepts voltages between 7 and 20 volts. It is also similar to the Arduino Nano and Leonardo. The Uno board and version 1.0 of Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer releases The Uno board is the first in a series of USB Arduino boards, and the reference model for the Arduino platform. The ATmega328 on the Arduino Uno comes preprogrammed with a bootloader that allows uploading new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol. The Uno also differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2

(Atmega8U2 up to version R2) programmed as a USB-to-serial converter Background

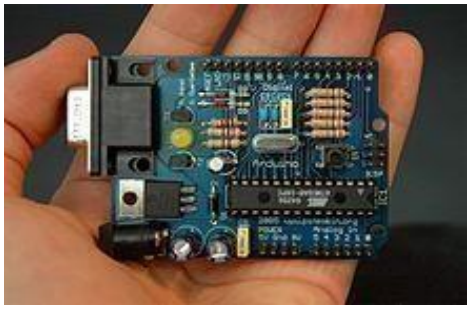


Fig 6.1: Arduino Controller

The Arduino project started at the Interaction Design Institute Ivrea (IDII) in Ivrea, Italy. At that time, the students used a BASIC Stamp microcontroller at a cost of $100, a considerable expense for many students. In 2003 Hernando Barragán created the development platform *Wiring* as a Master's thesis project at IDII, under the supervision of Massimo Banzi and Casey Reas, who are known for work on the Processing language. The Wiring platform consisted of a

printed circuit board (PCB) with an ATmega168 microcontroller, an IDE based on Processing and library functions to easily program the microcontroller.The cheaper ATmega8 microcontroller to Wiring. But instead of continuing the work on Wiring, they forked the project and renamed it *Arduino*. Early arduino boards used the FTDI USBtoserial driver chip and an ATmega168

Microcontroller: Microchip ATmega328P

- Operating Voltage: 5 Volt
- Input Voltage: 7 to 20 Volts
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- Analog Input Pins: 6
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50 mA
- Flash Memory: 32 KB of which 0.5 KB used by bootloader
- SRAM: 2 KB
- EEPROM: 1 KB
- Clock Speed: 16 MHz
- Length: 68.6 mm
- Width: 53.4 mm
- Weight: 25 g



Fig 6.2: Arduino UNO Pins General Pin

- LED: There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

- VIN: The input voltage to the Arduino/Genuino board when it's using an external power source. You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- 3V3: A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- GND: Ground pins.
- External Interrupts: pins 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- PWM(Pulse Width Modulation) 3, 5, 6, 9, 10, and 11 Can provide 8-bit PWM output with the analogWrite() function.
- TWI(Two Wire Interface): A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.
- AREF(Analog REFerence): Reference voltage for the analog inputs.[7]

Automatic (Software) Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino/Genuino Uno board is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip.[7]

Features

• High-performance, Low-power AVR® 8-bit Microcontroller

• Advanced RISC Architecture

– 131 Powerful Instructions – Most Single-clock Cycle Execution

– 32 x 8 General Purpose Working Registers

– Fully Static Operation

– Up to 16 MIPS Throughput at 16 MHz – On-chip 2-cycle Multiplier

In-System Programming by On-chip Boot Program True Read-While-Write Operation –

Programming Lock for Software Security

• JTAG (IEEE std. 1149.1 Compliant) Interface

- Boundary-scan Capabilities According to the JTAG Standard
- Programming of Flash, EEPROM, Fuses, and Lock Bits through the JTAG

• Peripheral Features

- Two 8-bit Timer/Counters with Separate Prescalers and Compare Modes
- One 16-bit Timer/Counter with Separate Prescaler, Compare Mode, and Capture Mode
- Real Time Counter with Separate Oscillator
- Four PWM Channels
- 8-channel, 10-bit ADC
- 2 Differential Channels with Programmable Gain at 1x, 10x, or 200x

– Byte-oriented Two-wire Serial Interface

– Programmable Serial USART

– Master/Slave SPI Serial Interface

– Programmable Watchdog Timer with Separate On-chip Oscillator

– On-chip Analog Comparator

• Special Microcontroller Features

– Power-on Reset and Programmable Brown-out Detection

– External and Internal Interrupt Sources

• Operating Voltages

– 2.7 - 5.5V for ATmega32L

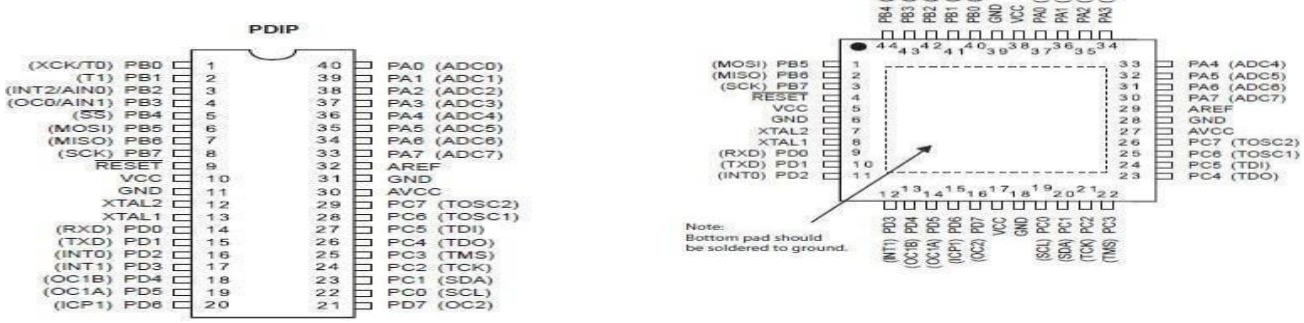
– 4.5 - 5.5V for ATmega32

• Power Consumption at 1 MHz, 3V, 25°C for ATmega32L

– Active: 1.1 mA

– Idle Mode: 0.35 mA

– Power-down Mode: < 1 µA



Pin Configurations AVR

CPU core

The ATmega32 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. By executing powerful instructions in a single clock cycle, the ATmega32 achieves throughputs approaching 1 MIPS per MHz allowing the system designer to optimize power consumption versus processing speed.

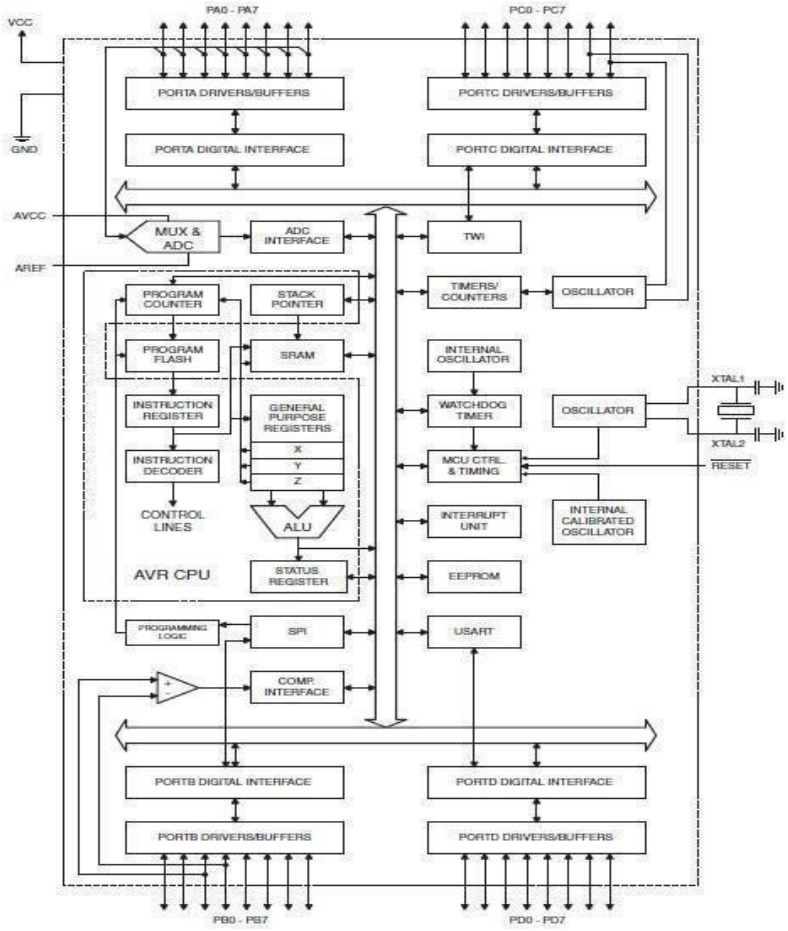


Fig 6.3: Internal Architecture GND Ground.

Port A (PA7..PA0) Port A serves as the analog inputs to the A/D Converter.

Port A also serves as an 8-bit bi-directional I/O port, if the A/D Converter is not used. Port B (PB7..PB0) Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit).

Port C (PC7..PC0) Port C is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit).

If the JTAG interface is enabled, the pull-up resistors on pins PC5(TDI), PC3(TMS) and

PC2(TCK) will be activated even if a reset occurs.

The TD0 pin is tri-stated unless TAP states that shift out data are entered.

Port C also serves the functions of the JTAG interface.

Port D (PD7..PD0) Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit).

RESET Reset Input. A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running

XTAL1 Input to the inverting Oscillator amplifier and input to the internal clock operating circuit.

XTAL2 Output from the inverting Oscillator amplifier.

AREF AREF is the analog reference pin for the A/D Converter. Registers AVR

Core

CHAPTER 7
HARDWARE COMPONENTS

7.1 BATTERY

A lithium-ion battery or Li-ion battery (abbreviated as LIB) is a type of rechargeable battery. Lithium-ion batteries are commonly used for portable electronics and electric vehicles and are growing in popularity for military and aerospace applications. In the batteries, lithium ions move from the negative electrode through an electrolyte to the positive electrode during discharge, and back when charging. Li-ion batteries use an intercalated lithium compound as the material at the positive electrode and typically graphite at the negative electrode. The batteries have a high energy density, no memory effect (other than LFP cells) and low selfdischarge. They can however be a safety hazard since they contain a flammable electrolyte, and if damaged or incorrectly charged can lead to explosions and fires.



Fig 7.1: BATTERY

Rather of merely utilizing this form of power supply units, a conventional power supply package may also be used, as seen below. Yet that would be a downside to the operation of a computer which should be considered with a handheld computer.

Description of Lithium-ion Battery

Machine Different types of machinery are used for Lithium-ion batteries manufacturing unit.

Which are as follow:

a) Sorting Machine

b) Grading Machine

c) CDS

Benefits of using Lithium-ion Batteries

Lithium-ion (Li-ion) batteries are inarguably the most popular type of rechargeable battery for consumer electronics. They can be used for a variety of products from mobile phones to cars, and their qualities are superior compared to other rechargeable batteries. At Night Searcher we use high-quality lithium-ion (Li-ion) batteries for all, they allow us to provide the high-performance, durable products our customers are used to.

7.2 Design of the robot

The frame of the vehicle is made up of Aluminum bars because of its lightweight and good strength, and the base is made up of Hylam sheet. The diameter of the wheels and the length of the rack are selected based on the ground clearance which is required to perform drilling mechanisms. The diameter of the funnel was chosen as 1cm to ensure that the system was able

to sow seeds of several sizes. Track belt is provided on the wheels of the robot and it helps to move on muddy terrains without sinking

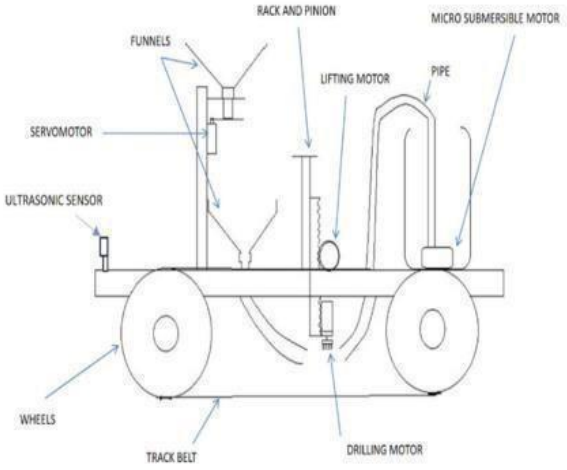


Fig7.2: Design of the robot

DC motors comprise of one collection of coils, or armature windings, between another wire coil or a series of magnetic materials, or the stator. Applying a tension to the coils creates a tension in the armature, resulting in movement. It consists of three major sections

1. Stator The Stator

2. Engine The Engine

3. Winding up

Construction function

DC motors comprise of one collection of coils, or armature windings, between another set of coils or a series of magnetic materials, or the stator. Applying a voltage to a coils creates a tension in the armature, resulting in motion.

The Stator

1. The stator is the fixed component of the engine outside.

2. The permanent magnet synchronous motor stator comprises of two or three magnetic field pole parts.

3. Additionally, the magnetic field may be produced by an electric motor. For this case, a DC coil (field winding) is wrapped around a magnet layer that is part of the stator.

The Engine

1. The rotor is the inner component that rotates.

2. The rotor consists of windings (called armature windings) which are attached to the external circuit by means of a mechanical connection.

3. Both the stator and the rotor are constructed of ferromagnetic steel. The two are divided by a distance in the sea.

Winding up

The winding consists of a series or a parallel link of the coils.

1. Armature winding-The winding by which the voltage is transmitted or caused.

2. Field winding-The coiling in which the present is transferred to create flux.

3. Windings are typically produced from copper.

Specifications

1. Voltage regulator Gear Engine BO(Battery Operated)

2. Volts: DC 3-9V

3. SPEED(RPM): 60RPM+-10% (no load)

4. Speed: continuous velocity

5. Form of mounting: horizontal or vertical **Principle of operation**

In the real DC motor, multiple such loops are bound to the rotor, both of which encounter committed to the growth to rotation. The greater the current in the wire, and the greater the gravitational pull, the quicker the wire travels because of the stronger power produced. At the same time as this torque is being generated. At various locations, the flux correlated with it varies, which allows the emf to be caused (e = d.. It voltage is in relation to the pressure that allows the current to pass the conductor which is alluded to it as a reverse voltage or emf.

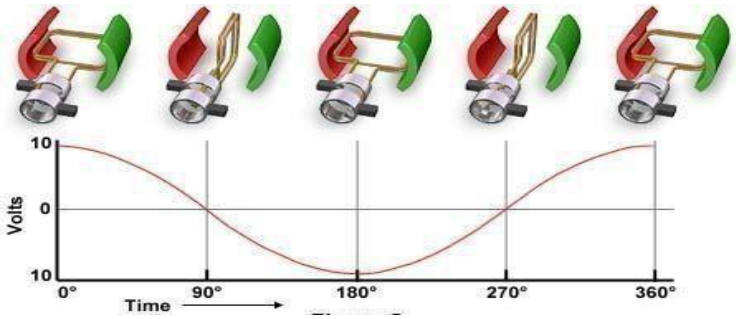


Fig7.3: Induced voltage in the armature winding of DC motor

Motor Driver L293D

The L293D is an embedded system engine driver that can be used for the synchronized, bi-directional operation of two small motors. Big, it means tiny. The L293D is restricted to 600 mA, but in fact it can only accommodate a couple of minor currents because you have done a significant heat sink to hold the case heat down, Hook up the circuit and power the motor while holding your eye on the button. You can't do it on the car because it's too heavy to handle .

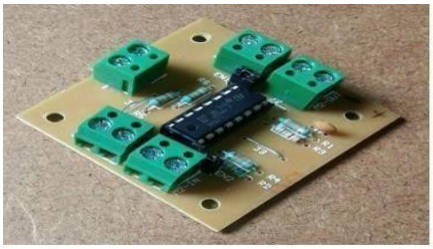


FIG 7.4: L293D Circuit

The L293 comes in a regular 16-pin, dual-in-line embedded system box. There's an L293 as well as an L293D component number. Use the "D" variant as it's installed in flyback diodes to eliminate inductive supply voltage.

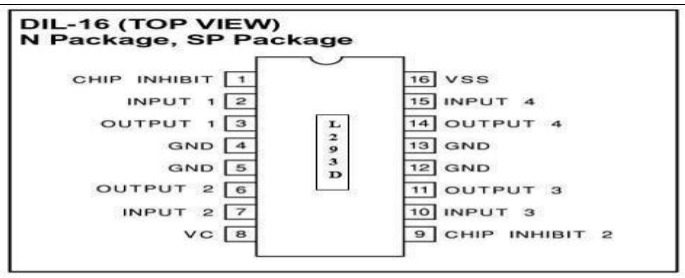


Fig 3.19: Pin out of L293D

2
9

3 D **L**

The following diagram illustrates how to attach the L293 to your engine and the Sticker. Growing engine has three Stamp pins. When you just have one generator, leave pins 9 , 10, 11 , 12, 13, 14, and 15 free.

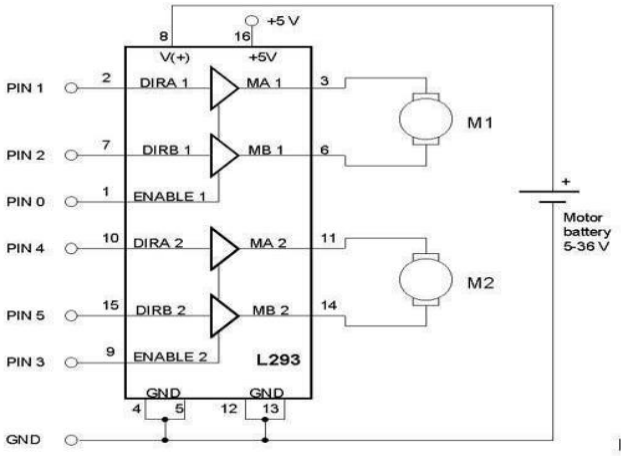


Fig 3.20: Internal diagram of L293D

Assume that you have just one engine linked to the allow connection to Stamp Pin 0, and the two steering controls are attached to Stamp Pin 1 & 2.

Hc-05

Communication device:-over project is based on wireless communication between micro controller and mobile phone. But alone micro controller is not able to communicate directly to the android mobile phone. Bluetooth Serial module's operation doesn't need drive, and can communicate with the other Bluetooth device that has the serial. But communication between two Bluetooth modules requires at

Least two conditions:

(1) The communication must be between master and slave.

(2) The password must be correct.

HC-05 module is an easy to use Bluetooth SPP (Serial Port Protocol) module, designed for transparent wireless serial connection setup. Serial port Bluetooth module is fully qualified Bluetooth V2.0+EDR (Enhanced Data Rate) 3Mbps Modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Blue core 04-External single chip Bluetooth system with CMOS technology and with AFH (Adaptive Frequency Hopping Feature).

Bluetooth Wireless networks for short range communications have a wide spread usage of Bluetooth radio transmissions between 2400–2480 MHz Modern mobile devices embed small, low-powered and cheap integrated chips functioning as short-range radio transceivers for Bluetooth radio communications. Device pairing, authentication, encryption and authorization techniques have given recognition to Bluetooth technology due to its vital security mechanisms.



Fig 7.5 :HC-05 Bluetooth

7.2.1. Specifications

Hardware features

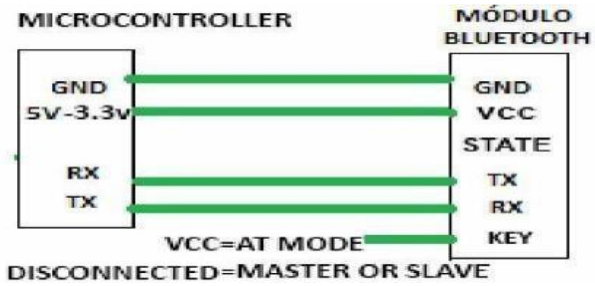
- Typical -80dBm sensitivity.
- Up to +4dBm RF transmits power.

- Low Power 1.8V Operation, 3.3 to 5 V I/O.
- With edge connector.

Software features

- Slave default Baud rate: 9600, Data bits:8, Stop bit:1,Parity:No parity.
- PIO9 and PIO8 can be connected to red and blue led separately. When master and slave are paired, red and blue led blinks 1time/2s in interval, while disconnected only blue led blinks 2times/s.
- Auto reconnect in 30 min when disconnected as a result of beyond the range of connection.

7.2.3. Typical Application Circuit:



How to get to the standard communication mode

1.Leave free KEY, don't connect it to VDD neither GND.

2.Supply power to the module. Then the module will enter to communication mode. It can be used for pairing.

CHAPTER 8

SOFTWARE REQUIREMENTS

The Arduino Software (IDE) is anything but difficult to-use for fledglings, yet sufficiently adaptable for cutting edge clients to exploit too. For instructors, it's helpfully in view of the

Processing programming condition, so understudies figuring out how to program in that condition will be acquainted with how the Arduino IDE functions.

Arduino is a model stage (open-source) in perspective of an easy to-use gear and programming. It includes a circuit board, which can be tweaked (suggested as a microcontroller) and a moment programming called Arduino IDE (Integrated Development Environment), which is used to make and exchange the PC code to the physical board.

The key highlights are:

1.Arduino sheets can read straightforward or propelled data signals from different sensors and change it into a yield, for instance, starting a motor, turning LED on/off, connect with the cloud and various distinctive exercises.

2. Unlike most past programmable circuit sheets, Arduino does not require an extra piece of gear (considered a product build) with a particular ultimate objective to stack another code onto the board. You can simply use a USB interface.

3. Additionally, the Arduino IDE uses a streamlined interpretation of C++, making it less requesting to make sense of how to program.

4. Finally, Arduino gives a standard edge factor that breaks the components of the scaled down scale controller into a more open package.

8.1 OPERATION DEMO:

STEP 1: INSTALL THE ARDUINO SOFTWARE (IDE)

Download the most recent variant from this page: http://arduino.cc/en/Main/Software

Next, continue with the establishment and please permit the driver establishment process

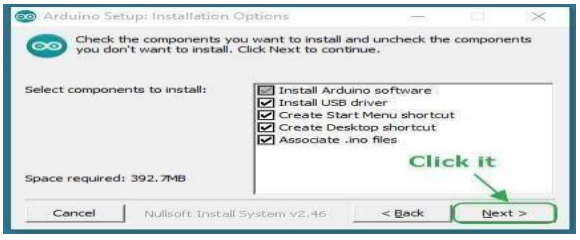
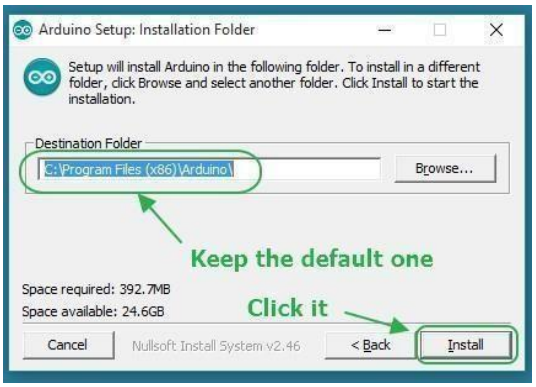


Fig.8.1.1 Arduino setup-installation options

Pick the parts to introduce and click "next" catch Pick the establishment index.



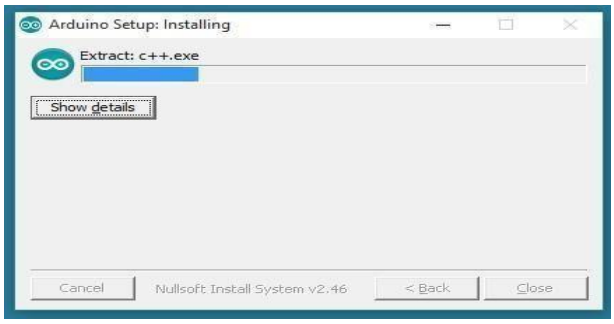


Fig .8.1.2 Arduino setup installation folder **Fig. 8.1.3Arduino setup-installing**

The procedure will separate and introduce all the expected documents to execute legitimately the Arduino Software (IDE)

STEP 2: GET AN UNO R3 AND USB CABLE

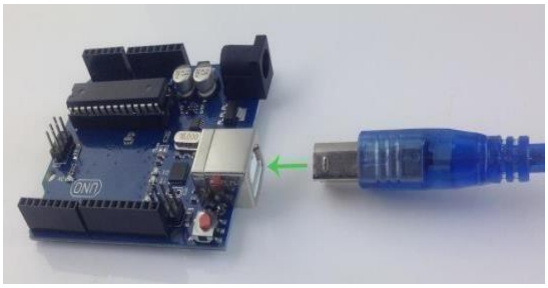
In this instructional exercise, you're utilizing an Uno R3. You additionally require a standard USB link (A fitting to B plug): the kind you would associate with a USB printer, for instance.



Fig. 8.1.4 USB cable UNO R3 board

STEP 3: CONNECT THE BOARD

The USB association with the PC is important to program The USB association with the PC is important to program the board and not simply to control it up. The Uno and Mega consequently draw control from either the USB or an outside power supply. Associate the board to your PC utilizing the USB link.



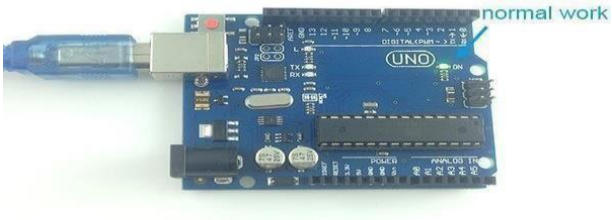
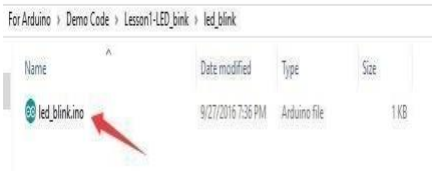


Fig. 8.1.5 Represents how to connect USB cable with UNO board

STEP 4: OPEN LESSON 1: LED BLINK

Open the LED blink example sketch: CD > For Arduino>Demo Code>Lesson1-LED_bink>led blink.



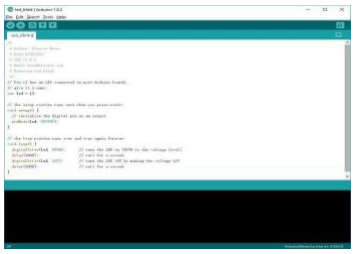


Fig. 8.1.6 LED valve and program to blink LED

STEP 5: SELECT YOUR BOARD

You'll need to select the entry in the Tools > Board menu that corresponds to your Arduino board.

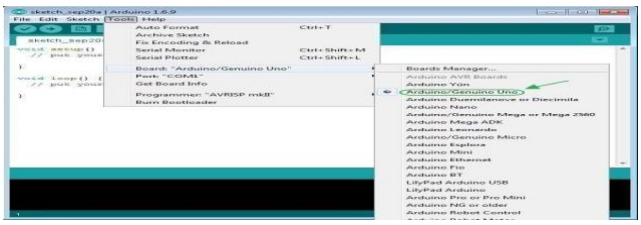


Fig. 8.1.7 path to select board

STEP 6: SELECT YOUR SERIAL PORT

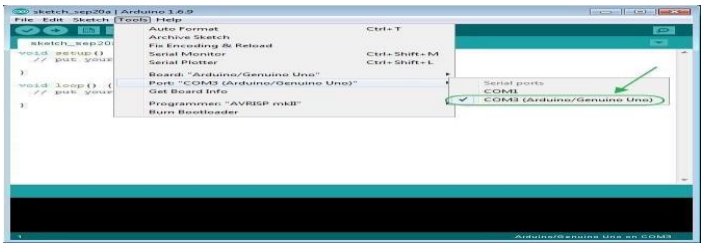


Fig. 8.1.8 path to select serial port

Select the serial gadget of the board from the Tools | Serial Port menu. This is probably going to be COM3or higher. To discover, you can detach your board and re-open the menu; the passage that vanishes ought to be Arduino board. Reconnect board and select that serial port .

STEP 7: UPLOAD THE PROGRAM

Presently, essentially tap the "Transfer" catch in the earth. Hold up a couple of moments - you should see the RX and TX leds on the board blazing.

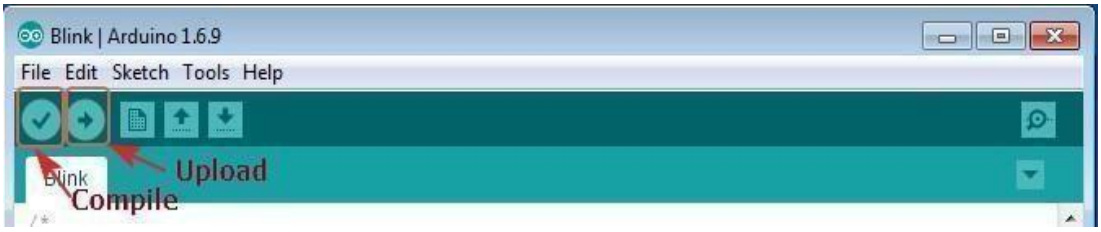


Fig. 8.1.9 path to upload

STEP 8: RESULT

8.2 ARDUINO INTERFACE INTRODUCTION

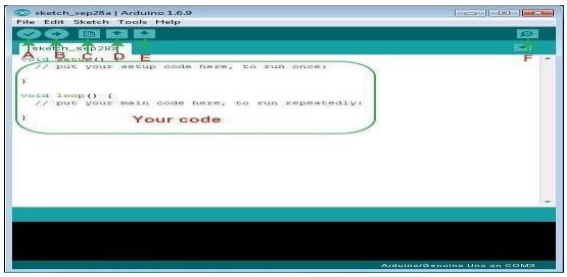


Fig. 8.2 Tools in Arduino interface software

8.3 ARDUINO UNO R3 HARDWARE INTRODUCTION



Fig. 8.3 UNO with hardware

8.4 HOW TO ADD LIBRARY FILES

Step 1:

Add library file: Sketch>Include Library>Add.ZIP Library

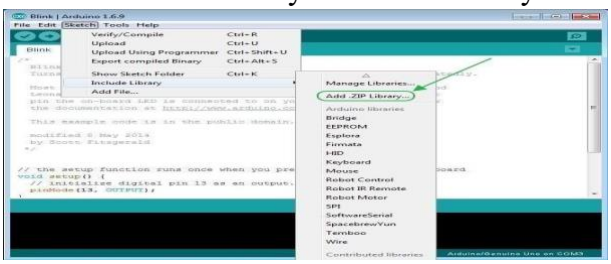


Fig. 8.4.1 path to add library files

STEP 2:

SELECT YOUR LIBRARY FILE COMPRESSION PACKAGE ON THE DEMO CODE

FILE, AS FOLLOWS:

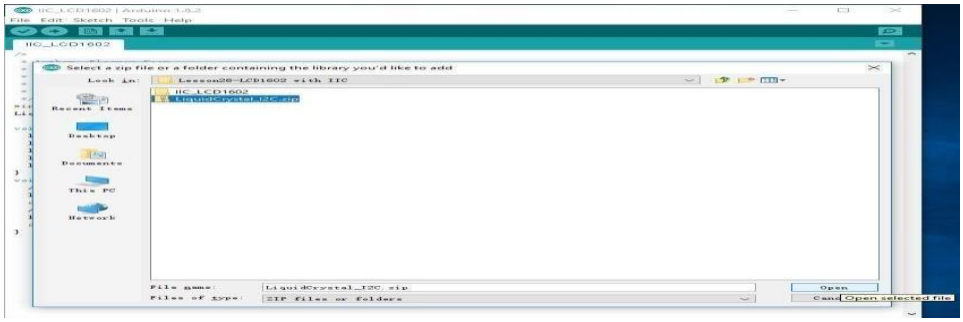


Fig .8.4.2 selecting the file to Arduino

8.5 BASIC COMMAND SETUP

For simplicity, the numbers could represent different actions such as moving forward, backward, turning, and stopping. For example:

- **Command "f"**: Move forward
- **Command "b"**: Move backward
- **Command "l"**: Turn left
- **Command "r"**: Turn right
- **Command "s"**: Stop

8.5.1 Using Bluetooth with Phone Commands

If using Bluetooth, your app would send the corresponding number string (like "1", "2", "3", "4") to the Arduino over Bluetooth. Here's how the logic could work:

- **Phone sends** "f" → moves forward.
- **Phone sends** "b" → moves backward.
- **Phone sends** "l" → turns left.
- **Phone sends** "r" → turns right
- **Phone sends** "s" → stop

Light Control Commands

- **Command "1"** – Light ON
- **Command "2"** – Light OFF

Buzzer Control Commands

- **Command "3"** – Buzzer ON
- **Command "4"** – Buzzer OFF

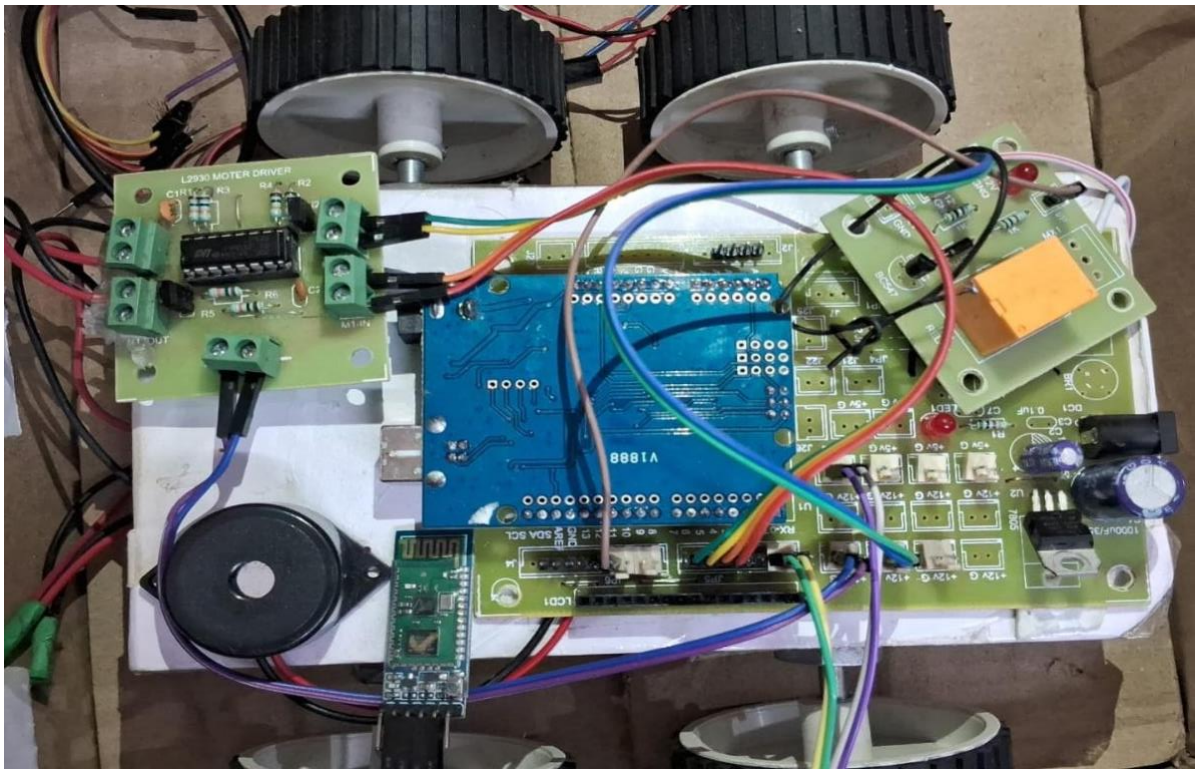
CHAPTER 9

RESULTS

The **smart wheelchair** is an innovative mobility system designed to assist individuals with physical disabilities by providing **voice - controlled navigation and safety features**. The operation of the wheelchair is based on **Bluetooth communication** between a **mobile application** and an **Arduino Uno microcontroller**.In addition to movement, the system integrates **headlight control** and a **buzzer mechanism**. To enhance user safety and convenience.The **buzzer** can also be **activated ordeactivated** to alert nearby individuals or signal emergency situations.

The integration of GPS in the **Voice Controlled Smart Wheelchair** enhances user safety by enabling real-time location tracking.

This combination of hardware and software ensures **smooth, responsive, and reliable control** of the wheelchair.



CHAPTER 10

CONCLUSION

Wireless controller is one of the most prominent basic needs for all living beings. But sadly, due to a huge amount of data and communication overheads the technology is not fully utilized. Many of the wireless-controlled robots use RF modules. The control commands open are more than RF modules. For this the android mobile user has to establish an application on her/his mobile. Then the user requires to switch on the Bluetooth in the mobile. The wireless connection method used to command the robot is Bluetooth technology. User can practice many commands like move forward, reverse, move left, move right using these commands which are transmitted from the Android mobile.

Robot has a Bluetooth transceiver unit that accepts the commands and return it to the microcontroller circuit to command the motors. Android Bluetooth-enable phones and Bluetooth module via HC-05 and information among Bluetooth tools. It is presumed that smart existence will slowly turn into a reality that user can control their home remotely and wirelessly

The smart wheelchair provides an efficient and user-friendly mobility solution for physically disabled individuals. Through voice commands sent via a mobile app and processed by Arduino Uno, the wheelchair's movement, lights, and buzzer are easily controlled. The **light feature** enhances visibility in dark environments, while the **buzzer** serves as a safety alert or warning system. Together, these functions ensure safer operation and greater independence. In the **Voice Controlled Smart Wheelchair Using GPS Tracker**, the GPS module plays a very important role in ensuring **safety and tracking** of the user. This project successfully demonstrates how technology can make mobility smarter, safer, and more convenient.

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